

Executive Summary:

Open Neighbourhoods Precede Concept Spread

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Headline Finding

When a scientific concept enters a new disciplinary subfield, the share of its initial co-occurrence partners that are native to that host subfield predicts subsequent uptake by newcomer authors. This *host-entry grafting* effect replicates on an independent biomedical population built from MeSH descriptors: **IRR/SD** = **1.23** [1.12, 1.36], **Holm** $p < 0.001$. The inverse-variance-weighted pooled estimate across three folds (screen, held-out, MeSH) is 1.26 [1.17, 1.36] with $I^2 = 0$.

Study Overview

This study investigates how emerging scientific concepts spread across disciplinary communities by analysing 426 semantically grounded concepts and 462,812 works from OpenAlex. Two research questions guide the work: (RQ1) which temporal and structural patterns characterise the emergence of scientific concepts, and (RQ2) how emerging concepts diffuse across disciplinary communities.

The study was conducted across five iterations: data assembly and pre-registration, network construction and precursor testing, deepening the closure lead and grafting screen, held-out confirmation and cross-population replication, and post-confirmation decomposition.

Key Results

Host-Entry Grafting (RQ2) — Confirmed and Replicated

When a concept first appears in a non-origin subfield, the host-nativeness of its entry partners (A_{cont}) predicts whether newcomer authors subsequently adopt it. Co-transfer of origin companions is null in every specification.

Table 1: Grafting results across folds (co-primary specification).

Fold	IRR/SD	95% CI	p	N
Screen	1.30	[1.16, 1.45]	10^{-5}	1,544
Held-out	1.19	[1.06, 1.33]	0.004	972
MeSH replication	1.23	[1.12, 1.36]	< 0.001	2,171
IVW pooled	1.26	[1.17, 1.36]	—	—

Post-confirmation decomposition (Iteration 5) shows the effect operates through both the extensive margin (IVW +6.43 pp/SD [4.76, 8.11]) and the intensive margin (IVW IRR/SD 1.183 [1.104,

1.267]). It is host-specific rather than reflecting general concept accessibility (IVW retention 0.969 [0.81, 1.10]) and does not vary by origin field (Wald $p = 0.638$).

Emergence Precursors (RQ1) — Partially Confirmed

The primary closure measure did not survive held-out confirmation (Holm $p = 0.147$). Only *persistent-neighbour closure*, restricted to top-20 associates present in consecutive years, is confirmed on the held-out fold ($S = -1.073$, 95% CI $[-1.86, -0.29]$, Holm $p = 0.015$). Burt constraint is positive on both folds, ruling out a simple brokerage explanation. Precursors add no predictive value for *whether* a concept will show sustained uptake.

Diffusion Typology and Ordering

A $k = 2$ typology separates localised concepts ($n = 136$) from concepts that are broad from the start ($n = 66$), replicating on the held-out fold. Network expansion precedes disciplinary diffusion in 25 of 26 pooled main-arm dual-onset concepts.

Adopter-Level Mechanism

Adopters are 3.09 times more likely than matched non-adopters to have prior exposure to the concept’s entry partners (OR 3.09 [2.32, 4.28]). Enrichment is strongest for origin-vocabulary (FOREIGN) partners (OR 2.88), running in the opposite direction from the entry-level grafting effect.

Dead Ends and Null Results

- **Source-sink viability:** blocked by insufficient citation-layer density (Gate A failure, 17.3% within-host traced share vs the 40% threshold).
- **Prediction of emergence:** precursors add no value beyond frequency and centrality baselines ($\Delta\text{AUC} -0.010 [-0.045, 0.021]$).
- **Breadth prediction from openness:** null on both screen and held-out folds.
- **Closure–anchoring link:** no detectable association between pre-emergence closure and later host-entry anchoring.
- **Co-transfer:** null across all folds and populations.

Open Questions

1. The population is dominated by physics, computer science, and biomedicine; generalisation to social sciences and humanities is untested.
2. The primary fully-saturated specification is underpowered (held-out: 250 events, 30 clusters).
3. Whether the host-vocabulary gradient operates through genuine translation of ideas into host vocabulary or through artefacts of topic-classification overlap.
4. The absorptive-capacity vs topical-proximity ambiguity in the adopter mechanism.

Target venue: Applied Network Science.